Digital Image Processing Case Study Solutions

(Structured according to university exam/assignment rubric with concept, application, justification, and processing sequence.)

1. Dental X-Ray Enhancement and Cavity Region Extraction

Problem Analysis

Dental X-ray images generally suffer from:

- Low contrast
- Quantum noise
- Dark appearance
- Small cavity regions hidden inside enamel and dentine
- Unclear cavity boundaries

Processing Pipeline

Step 1: Grey-Level Enhancement

(a) Image Negative Transformation

$s=L-1-rs=L-1-r$

where:

- rr = input pixel value
- ss = transformed pixel value

Purpose:

- Cavities appear as dark regions.
- Negative transformation converts them into bright regions.
- Improves visual distinction between healthy and decayed tissue.

(b) Power-Law (Gamma) Transformation

$s=cr\gamma s=cr^\gamma$

Choose:

$\gamma<1\gamma < 1$

Reason:

- Enhances darker regions.
- Reveals early-stage cavities hidden within low-intensity areas.

Step 2: Histogram Equalization

Purpose

- Redistributes intensity values.
- Enhances global contrast.
- Makes cavity regions more distinguishable from surrounding enamel.

Result:

- Dark cavity pixels become visually separable from normal tooth structure.

Step 3: Noise Removal

Median Filter (3×3 or 5×5)

$Median(x,y)Median(x,y)$

Why Median Filter?

- Removes salt-and-pepper and quantum noise.
- Preserves cavity boundaries.
- Better than averaging filter, which may blur cavity edges.

Step 4: Segmentation

Thresholding

Select threshold TT

$$g(x,y)=\{1,f(x,y)>T0,f(x,y)\leq Tg(x,y)= \begin{cases} 1, & f(x,y)>T\\ 0, & f(x,y) \le T \end{cases}$$

Purpose

- Separates cavity pixels from healthy tissue.

Alternative: Region Growing

Procedure:

1. Select cavity seed point.
2. Add neighboring pixels of similar intensity.
3. Grow region until threshold criterion fails.

Advantage

- Captures irregular cavity shapes.

Step 5: Edge Detection

Canny Edge Detector

Operations:

1. Gradient computation
2. Non-maximum suppression
3. Hysteresis thresholding

Purpose

- Produces accurate cavity boundaries.
- Less sensitive to noise.

Final Sequence

```
Dental X-ray
↓
Negative / Gamma Transformation
↓
Histogram Equalization
↓
Median Filtering
↓
Thresholding or Region Growing
↓
Canny Edge Detection
↓
Extracted Cavity Region
```

Justification

Technique	Purpose
Negative Transformation	Highlights dark cavities
Gamma Correction	Enhances low-intensity details
Histogram Equalization	Improves overall contrast
Median Filter	Removes noise while preserving edges
Thresholding/Region Growing	Isolates cavity region
Canny Edge Detection	Precisely outlines cavity

2. Night-Time Vehicle Number Plate Region Extraction

Challenges

- Very low brightness
- Headlight glare

- Motion blur
- Sensor noise
- Low visibility of characters

Processing Pipeline

Step 1: Brightness Enhancement

Log Transformation

$s=clog⁡(1+r)s=c|log(1+r)$

Purpose

- Expands dark pixel values.
- Reveals details hidden in shadows.

Step 2: Histogram Processing

Contrast-Limited Adaptive Histogram Equalization (CLAHE)

Advantages

- Enhances dark areas.
- Prevents over-amplification of headlight glare.
- Produces local contrast improvement.

Step 3: Noise Reduction

Median Filter

$3×33|times3$

Purpose

- Eliminates sensor noise.
- Preserves plate edges.

Step 4: Sharpening

Unsharp Masking

$fs=f+k(f−fblur)f_s=f+k(f-f_{blur})$

or

Laplacian Filter

$∇2f\nabla^2f$

Purpose

- Restores blurred character boundaries.
- Improves edge sharpness.

Step 5: Edge Detection

Canny Edge Detection

Detect:

- Plate border
- Character edges

Step 6: Line Detection

Hough Transform

Detects:

- Horizontal edges
- Vertical edges

forming a rectangular plate.

Step 7: Edge Linking

Purpose:

- Connect broken plate boundaries.
- Form closed rectangle.

Step 8: Character Segmentation

Adaptive Thresholding

Separates:

- Characters
- Background

even under uneven illumination.

Final Sequence

```
Night Image
↓
Log Transformation
↓
CLAHE
↓
Median Filter
↓
Unsharp Masking / Laplacian
↓
Canny Edge Detection
↓
Hough Line Detection
↓
Edge Linking
↓
Adaptive Thresholding
↓
```

```
Plate and Character Extraction
```

Justification

Technique	Contribution
Log Transformation	Enhances dark regions
CLAHE	Controls glare while improving visibility
Median Filter	Removes noise
Unsharp Masking	Sharpens characters
Canny	Detects strong edges
Hough Transform	Detects plate rectangle
Adaptive Thresholding	Segments characters

3. Textile Fabric Defect Detection Using Frequency-Domain Processing

Challenges

- Repetitive weave texture
- Periodic lighting noise
- Uneven illumination
- Small defects hidden in texture

Processing Pipeline

Step 1: Fourier Transform

Convert image to frequency domain.

$F(u,v)F(u,v)$

Purpose:

- Separate periodic components from defects.

Step 2: Notch/Band-Reject Filtering

Remove periodic peaks corresponding to:

- Lighting noise
- Regular weave pattern

Result:

- Fabric texture suppressed.
- Defects become prominent.

Step 3: Homomorphic Filtering

Used to correct uneven illumination.

$Image = Illumination \times Reflectance$

Enhances reflectance while reducing illumination variation.

Step 4: High-Pass Filtering

Enhances abrupt changes caused by:

- Holes
- Knots
- Oil stains
- Broken threads

Step 5: Inverse Fourier Transform

Convert filtered image back to spatial domain.

Step 6: Thresholding

Segment defect pixels.

Step 7: Boundary Detection

Apply:

- Canny edge detector
- Morphological boundary extraction

to outline defective areas.

Final Sequence

```
Fabric Image
↓
FFT
↓
Notch/Band-Reject Filtering
↓
Homomorphic Filtering
↓
High-Pass Filtering
↓
Inverse FFT
↓
Thresholding
↓
Boundary Detection
↓
Defect Extraction
```

Why Frequency Domain?

Because the weave pattern and lighting noise are periodic.

Frequency-domain filtering:

- Removes repetitive components efficiently.
- Requires fewer operations than large spatial filters.
- Better isolates irregular defects.

4. Blood Cell Segmentation in Microscopic Images

Challenges

- Uneven staining
- Overlapping cells
- Dust particles
- Low contrast

Processing Pipeline

Step 1: Contrast Improvement

Adaptive Histogram Equalization (CLAHE)

Purpose:

- Enhances local cell contrast.
- Improves separation of cells and background.

Step 2: Noise Removal

Median Filter

or

Morphological Opening

Removes:

- Dust particles
- Small artifacts

while preserving cell boundaries.

Step 3: Thresholding

Adaptive Thresholding

Preferred because staining varies throughout image.

Produces:

- Binary cell image

Step 4: Region-Based Segmentation

Region Growing

Procedure:

1. Select seed cell.
2. Expand to neighboring pixels.
3. Stop at boundaries.

OR

Region Splitting and Merging

Useful for overlapping cells.

Step 5: Cell Separation

Watershed Transformation

Separates touching cells based on distance maps.

Step 6: Edge Detection

Canny Edge Detection

Detects individual cell contours.

Step 7: Boundary Detection

Extract complete boundary for:

- Cell counting
- Shape measurement

Final Sequence

```
Microscopic Image
↓
CLAHE
↓
Median Filter
↓
Adaptive Thresholding
↓
Region Growing / Splitting-Merging
```

```
↓
Watershed Separation
↓
Canny Edge Detection
↓
Boundary Detection
↓
Individual Cell Regions
```

Justification

Technique	Function
CLAHE	Enhances cell visibility
Median Filter	Removes dust noise
Adaptive Thresholding	Handles uneven staining
Region Growing	Extracts cell regions
Watershed	Separates overlapping cells
Canny	Detects cell boundaries

5. Text Extraction from Degraded Historical Documents

Challenges

- Yellowed pages
- Ink bleed-through
- Stains
- Uneven illumination
- Faded handwriting

Processing Pipeline

Step 1: Contrast Enhancement

Gamma Correction

$\gamma < 1 \backslash gamma < 1$

Enhances faded strokes.

Step 2: Histogram Processing

Histogram Equalization

or

CLAHE

Purpose:

- Separates text from paper background.
- Enhances faint writing.

Step 3: Noise and Stain Removal

Median Filter

Removes:

- Small stains
- Scanning artifacts

without damaging handwriting.

Step 4: Stroke Enhancement

Unsharp Masking

Strengthens pen strokes.

$fs=f+k(f−fblur)f_s=f+k(f-f_{blur})$

Step 5: Thresholding

Comparison

Otsu's Global Thresholding

Advantages:

- Simple
- Automatic

Disadvantages:

- Fails with uneven lighting.
- Cannot handle yellowed backgrounds effectively.

Adaptive Thresholding (Preferred)

Advantages:

- Calculates threshold locally.
- Handles illumination variation.
- Preserves faded handwriting.

Chosen Method: Adaptive Thresholding

Step 6: Edge Detection

Canny Edge Detector

Detects:

- Character boundaries
- Word edges

Step 7: Boundary Detection

Identify:

- Text lines
- Individual word regions
- Character groups

using connected-component analysis.

Final Sequence

```
Historical Document
↓
Gamma Correction
↓
CLAHE / Histogram Equalization
↓
Median Filtering
↓
Unsharp Masking
↓
Adaptive Thresholding
↓
Canny Edge Detection
↓
Boundary Detection
↓
Text Region Extraction
```

Justification

Technique	Reason
Gamma Correction	Enhances faded ink
CLAHE	Improves local contrast
Median Filter	Removes stains
Unsharp Masking	Strengthens handwriting
Adaptive Thresholding	Handles uneven illumination
Canny Edge Detection	Detects text boundaries
Boundary Detection	Extracts words and text lines

Conclusion

For all five applications, the recommended Digital Image Processing workflow follows:

Image Acquisition → Gray-Level Enhancement → Histogram Processing → Noise Reduction → Sharpening (if required) → Segmentation → Edge Detection → Boundary Extraction → Final Object/Region Detection

This sequence maximizes visibility, minimizes noise, preserves important boundaries, and produces accurate segmentation results suitable for practical medical, transportation, industrial, biological, and document-processing applications.